CSCE 548 — Project 4

Calorie Tracker — Deployment & System Test Document

Project Overview

The Calorie Tracker is an n-tier web application for logging daily calorie intake and tracking food items. Users can create accounts, log meals with nutritional data, and manage individual food items per meal entry. The system is built across three tiers: a MySQL database, a Spring Boot REST API, and a plain HTML/CSS/JS frontend.

Layer	Technology	Default Port
Database (Data Layer)	MySQL 8.x	3306
REST API (Business + Service)	Spring Boot 4 / Java 17	8080
Frontend (Client Layer)	HTML + CSS + JavaScript	Any browser / static server

1. Prerequisites

Install the following before setup:

Java 17 JDK — Download from https://adoptium.net. Verify with: java -version

MySQL 8.x — Download from https://dev.mysql.com/downloads/. Note your root password during setup.

Git — Download from https://git-scm.com

A modern web browser — Chrome, Firefox, or Edge to run the frontend.

2. Download the Repository

Clone the repo or download the ZIP from GitHub:

```
git clone https://github.com/Rishapatel09/CSCE548.git
```

```
cd CSCE548
```

3. Set Up the Database

Open a terminal and run both SQL scripts:

```
mysql -u root -p < db_creation.sql
```

```
mysql -u root -p < db_inserts.sql
```

This creates the calorie_tracker database with three tables (app_user, calorie_entry, food_item) and inserts sample data.

4. Configure the Backend

The database credentials are set in two places. Both must match your MySQL installation:

File 1: calorie-tracker-api/src/main/resources/application.properties

```
spring.application.name=calorie-tracker-api
```

```
spring.jackson.deserialization.fail-on-null-for-primitives=false
```

File 2: calorie-tracker-api/src/main/java/com/calorietracker/repository/DBUtil.java

```
private static final String USER = "root";
```

```
private static final String PASS = "your_mysql_password";
```

Update the PASS value in DBUtil.java to match your MySQL root password.

5. Build and Run the Backend

Navigate into the API folder and run:

```
cd calorie-tracker-api
```

```
./gradlew bootRun # macOS / Linux
```

```
gradlew.bat bootRun # Windows
```

Wait for: Started CalorieTrackerApiApplication in the console. Verify by visiting http://localhost:8080/users in your browser — you should see a JSON list of users.

6. Open the Frontend

Open Frontend/index.html directly in your browser. No build step required. The app connects to http://localhost:8080 by default. You can change the server address in the API Settings section if needed.

7. API Endpoints

Method	Endpoint	Description
GET	/users	Get all users
GET	/users/{id}	Get user by ID
POST	/users	Create new user
PUT	/users/{id}	Update user

DELETE	/users/{id}	Delete user
GET	/entries	Get all calorie entries
GET	/entries/{id}	Get entry by ID
GET	/users/{id}/entries	Get entries for a user
POST	/entries	Create calorie entry
PUT	/entries/{id}	Update calorie entry
DELETE	/entries/{id}	Delete calorie entry
GET	/food-items	Get all food items
GET	/food-items/{id}	Get food item by ID
GET	/food-items/entry/{id}	Get food items for an entry
POST	/food-items	Create food item
PUT	/food-items/{id}	Update food item
DELETE	/food-items/{id}	Delete food item

8. Full System Test

All CRUD operations were tested using the frontend at Frontend/index.html with the backend running at http://localhost:8080. Tests were performed for all three entities: Users, Calorie Entries, and Food Items.

Users

Show All Users: Click 'Show All Users'. Table populates with all user records from the database.

Find a User: Enter a user ID and click 'Find User'. Returns name, email, and created date.

Add New User: Enter a name and email, click 'Create User'. Returns confirmation with new user details.

Edit User: Enter user ID, new name and email, click 'Update User'. Returns success confirmation.

Remove User: Enter user ID, click 'Remove User', confirm the dialog. User is deleted from database.

Calorie Entries

Show All Entries: Click 'Show All Entries'. Table shows all meal entries with date, meal type, and macros.

Entries for a User: Enter a user ID, click 'Find Entries'. Returns all entries belonging to that user.

Log a Meal: Fill in user ID, date, time, meal type, and nutritional values. Click 'Log Entry'. Entry is saved.

Edit an Entry: Enter entry ID and updated values, click 'Update Entry'. Returns success confirmation.

Remove an Entry: Enter entry ID, click 'Remove Entry'. Entry and its food items are deleted.

Food Items

Show All Food Items: Click 'Show All Food Items'. Table shows all foods with serving info and macros.

Food Items in a Meal: Enter an entry ID, click 'Find Food Items'. Returns all foods in that entry.

Add a Food Item: Fill in entry ID, food name, serving size, and nutritional values. Click 'Add Food Item'.

Edit a Food Item: Enter food ID and updated values, click 'Update Food Item'. Returns success.

Remove a Food Item: Enter food ID, click 'Remove Food Item'. Food item is deleted from database.

9. Changes Made to AI-Generated Code

• AppUser.java — changed userId field from primitive int to Integer (wrapper type) to allow null deserialization from JSON

• application.properties — added spring.jackson.deserialization.fail-on-null-for-primitives=false

• Frontend/index.html — completely rebuilt with user-friendly forms for all CRUD operations, replacing the raw JSON dashboard

• Frontend — removed all technical HTTP method badges (GET/POST/PUT/DELETE) and replaced with plain English labels

• Frontend — results display as formatted readable text instead of raw JSON

10. Troubleshooting

Connection refused on port 8080: The Spring Boot backend is not running. Start it with gradlew.bat bootRun.

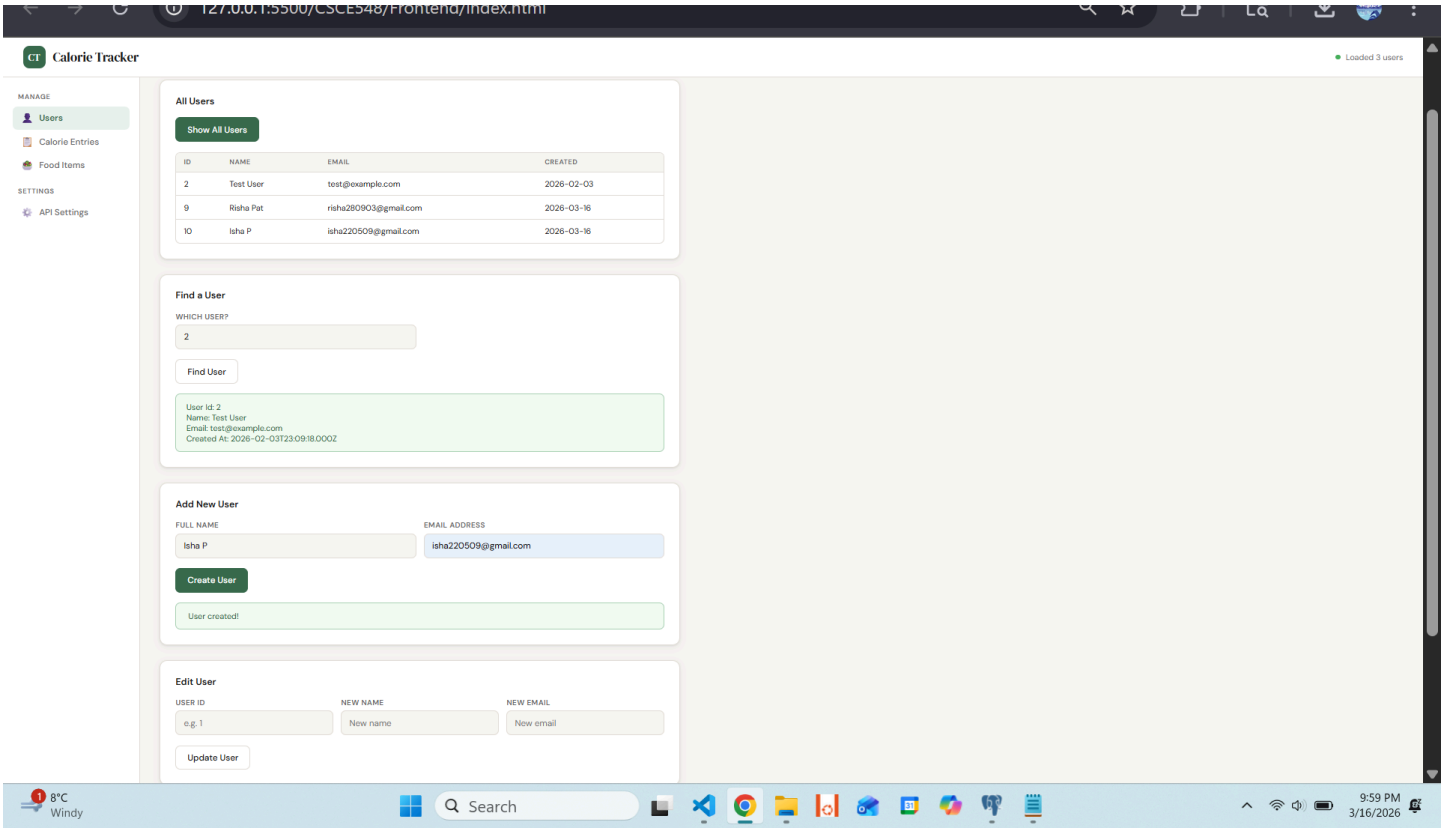
400 Bad Request on create/update: Check AppUser.java — userId must be Integer not int. Also verify application.properties has the Jackson config line.

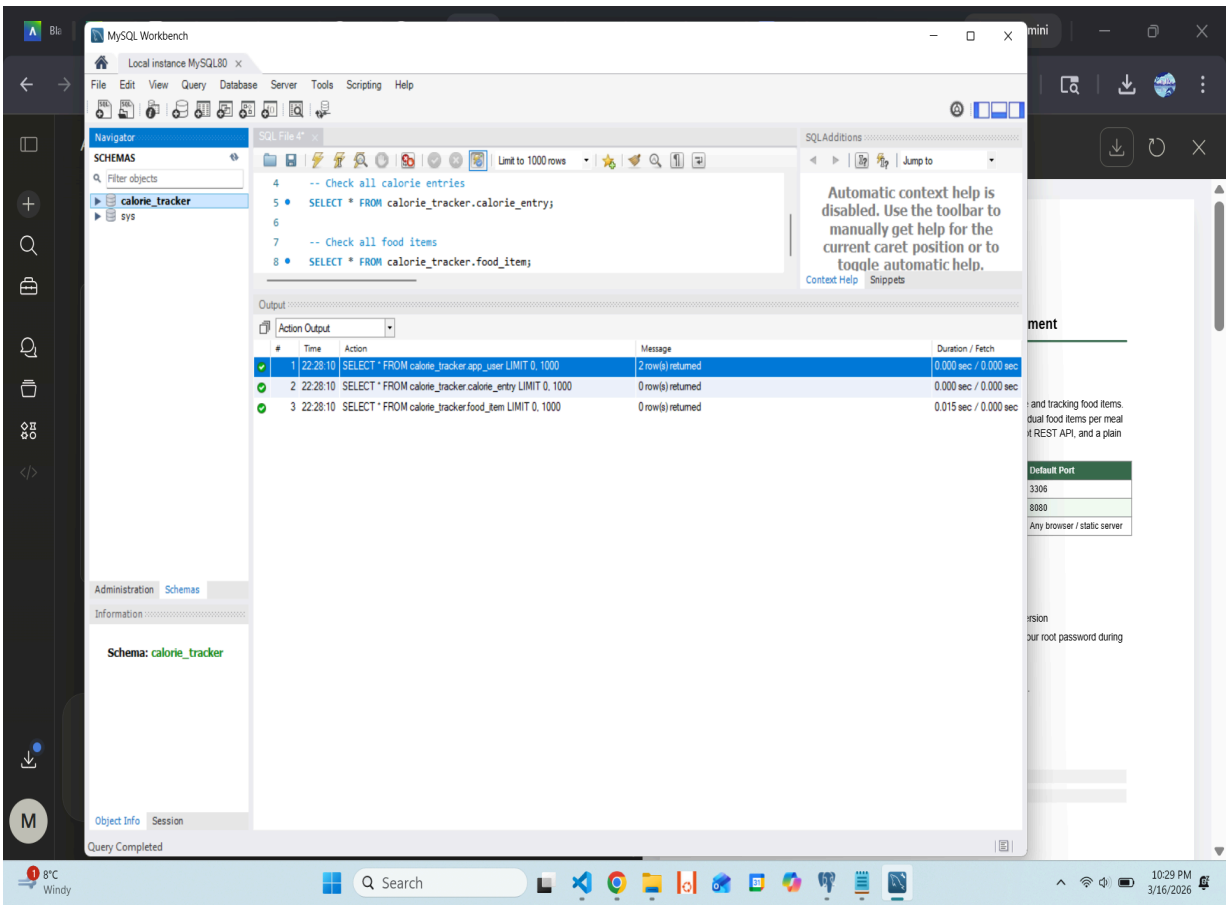
MySQL access denied: Update the PASS field in DBUtil.java to match your MySQL password.

Tables not found error: Re-run db_creation.sql in MySQL to recreate the schema.

gradlew: Permission denied (Mac/Linux): Run: chmod +x gradlew before running ./gradlew bootRun

Frontend shows Bad Request: Make sure the backend is fully started before submitting any forms.


127.0.0.1:5500/CSCE548/Frontend/index.html
Calorie Tracker
Loaded 3 users
MANAGE
Users
Calorie Entries
Food Items
SETTINGS
API Settings
All Users
Show All Users
ID NAME EMAIL CREATED
2 Test User test@example.com 2026-02-03
9 Risha Pat risha280903@gmail.com 2026-03-16
10 Isha P isha220509@gmail.com 2026-03-16
Find a User
WHICH USER?
2
Find User
User Id: 2
Name: Test User
Email: test@example.com
Created At: 2026-02-03T23:09:18.000Z
Add New User
FULL NAME
Isha P
EMAIL ADDRESS
isha220509@gmail.com
Create User
User created!
Edit User
USER ID
e.g. 1
NEW NAME
New name
NEW EMAIL
New email
Update User
8°C
Windy
Search
9:59 PM
3/16/2026


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
Navigator
SCHEMAS
Filter objects
calorie_tracker
sys
SQL File 4*
Limit to 1000 rows
-- Check all calorie entries
SELECT * FROM calorie_tracker.calorie_entry;
-- Check all food items
SELECT * FROM calorie_tracker.food_item;
SQL Additions
Jump to
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.
Context Help Snippets
Output
Action Output
Time Action Message Duration / Fetch
1 22:28:10 SELECT * FROM calorie_tracker.app_user LIMIT 0, 1000 2 row(s) returned 0.000 sec / 0.000 sec
2 22:28:10 SELECT * FROM calorie_tracker.calorie_entry LIMIT 0, 1000 0 row(s) returned 0.000 sec / 0.000 sec
3 22:28:10 SELECT * FROM calorie_tracker.food_item LIMIT 0, 1000 0 row(s) returned 0.015 sec / 0.000 sec
Administration Schemas
Information
Schema: calorie_tracker
Object Info Session
Query Completed
Default Port
3306
8080
Any browser / static server
8°C
Windy
Search
10:29 PM
3/16/2026

127.0.0.1:5500/CSCE548/Frontend/index.html

Calorie Tracker

Loaded 3 users

MANAGE

- Users
- Calorie Entries
- Food Items

SETTINGS

- API Settings

Show All Users

ID	NAME	EMAIL	CREATED
2	Austin	apa3941@gmail.com	2026-02-03
9	Risha Pat	risha280903@gmail.com	2026-03-16
10	Isha P	isha220509@gmail.com	2026-03-16

Find a User

WHICH USER?

2

Find User

User Id: 2
Name: Test User
Email: test@example.com
Created At: 2026-02-03T23:09:18.000Z

Add New User

FULL NAME: Isha P

EMAIL ADDRESS: isha220509@gmail.com

Create User

User created!

Edit User

USER ID: 2

NEW NAME: Austin

NEW EMAIL: apa3941@gmail.com

Update User

User updated successfully!

8°C Windy | Search | 10:00 PM 3/16/2026

127.0.0.1:5500/CSCE548/Frontend/index.html

Calorie Tracker

Loaded 2 users

MANAGE

- Users
- Calorie Entries
- Food Items

SETTINGS

- API Settings

Show All Users

ID	NAME	EMAIL	CREATED
9	Risha Pat	risha280903@gmail.com	2026-03-16
10	Isha P	isha220509@gmail.com	2026-03-16

Find a User

WHICH USER?

e.g. 1

Find User

Add New User

FULL NAME: e.g. Jane Smith

EMAIL ADDRESS: e.g. jane@example.com

Create User

Edit User

USER ID: e.g. 1

NEW NAME: New name

NEW EMAIL: New email

Update User

Remove User

USER ID

2

Delete User

User deleted successfully!

8°C Windy | Search | 10:01 PM 3/16/2026

127.0.0.1:5500/CSCE548/Frontend/index.html

CT Calorie Tracker

Found 1 entries

MANAGE

- Users
- Calorie Entries
- Food Items

SETTINGS

- API Settings

Calorie Entries

Track what you eat each day

All Entries

Show All Entries

ID	USER	DATE	TIME	MEAL	CAL	P	C	F
12	10	2026-03-18T04:00:00.000Z	15:00:00	Lunch	100	90	34	23

Entries for a User

WHICH USER?

10

Find Entries

Found 1 entry for this user.

Log a Meal

WHICH USER? e.g. 1

DATE mm/dd/yyyy

TIME --:-- --

8°C Windy

Search

10:03 PM 3/16/2026

127.0.0.1:5500/CSCE548/Frontend/index.html

CT Calorie Tracker

Loading entries...

MANAGE

- Users
- Calorie Entries
- Food Items

SETTINGS

- API Settings

Edit an Entry

ENTRY ID e.g. 1

WHICH USER? e.g. 1

DATE mm/dd/yyyy

TIME --:-- --

MEAL TYPE Select meal...

CALORIES e.g. 500

PROTEIN (G) e.g. 35

CARBS (G) e.g. 50

FATS (G) e.g. 18

NOTES Updated notes...

Update Entry

Remove an Entry

ENTRY ID

12

Delete Entry

Entry deleted successfully!

8°C Windy

Search

10:04 PM 3/16/2026